Technical Report: Threat Reconnaissance & Deception Analysis

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Focus Area: Threat Reconnaissance, Payload Obfuscation & SOC Detection Fundamentals

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1. Executive Summary

This report documents an educational proof-of-concept (PoC) exercise focused on basic threat reconnaissance, link obfuscation, and detection mechanisms within a Security Operations Center (SOC) framework. The analysis examines how basic browser-based IP logging provides initial reconnaissance data, how attackers structure multi-stage phishing campaigns using standard HTML, and how defensive controls—such as SIEM correlation rules and Endpoint Detection and Response (EDR) systems—identify these behavioral patterns.

2. Reconnaissance & Data Extraction (Educational PoC)

2.1 Test Environment Setup

- **Tool Used:** SnifferIP (snifferip.com) — an online utility used for generating tracking links to observe browser fingerprinting in a controlled lab setting.
- **Execution Scenario:** A tracking link was accessed via a personal mobile device to simulate a user clicking a link in a phishing message.

2.2 Captured Technical Data

Upon accessing the link, the following metadata was recorded in the control panel:

Field	Captured Data	Technical Interpretation
IP Address	196.154.247.240	Public IP provided by the ISP.
ISP	Vodafone Data	Mobile service provider.
Platform / OS	Android 10	Target operating system.
Browser	Chrome 153.0	Client browser version.
Location	Ash-Shaykh Zāyid / Giza	Approximate location based on ISP routing.
Fingerprint ID	1iv5no5	Unique hardware/browser signature.

2.3 Offensive Use Cases (Red Team Perspective)

- **Targeted Spear-Phishing:** Knowledge of the provider (Vodafone Data) and approximate region (Giza) enables attackers to craft context-aware social engineering lures.

- **Vulnerability Mapping:** Matching browser versions (Chrome 153.0 on Android 10) against public databases (NVD/CVE) helps identify potential unpatched vulnerabilities.

3. Link Obfuscation & Multi-Stage Redirection

To bypass basic email security filters that inspect plain URLs, multi-stage interaction models can be constructed:

- **HTML Button Redirection:** Creating a multi-page HTML sequence where the user must click an internal button (e.g., "Next Page" or "Page 2") to proceed.

- **Mechanism:** The initial landing page appears benign to simple scanner. The secondary interaction triggers the download or connects to an external payload destination.

- **User Interaction Obligation:** Forcing user interaction builds a chain of trust and bypasses non-interactive automated URL crawlers.

4.SOC Detection & Defensive Strategy (Blue Team Perspective)

While obfuscation techniques mask initial links, security monitoring focuses on **Behavioral Traces**:

- **Proxy & DNS Logs:** Web proxies monitor outbound HTTP/HTTPS requests, tracking redirection chains and flagging connections to newly registered domains (NRDs).

- **EDR Process Monitoring:** If an HTML page or browser process initiates secondary execution routines (e.g., spawning powershell.exe or cmd.exe), Endpoint Detection and Response tools isolate the process immediately.

- **SIEM Correlation Rules:** SIEM platforms aggregate logs across network controls and endpoints. A correlation rule flags events where an external link visit is immediately followed by abnormal process generation or file drops.

5. Deception Technologies (Honeybots)

- **Concept:** Deploying non-production, decoy assets (e.g., an internal router admin portal decoy) within the network environment.

- **Objective:** Since no legitimate user has a reason to interact with decoy assets, any access attempt triggers an immediate high-priority alert.

- **Outcome:** Captures attacker IP addresses and browser fingerprints early in the reconnaissance phase, enabling proactive firewall blocking before sensitive systems are reached.

6. Conclusion & Key Takeaways

- **Reconnaissance Limits:** IP logging provides environmental context for social engineering, but does not constitute direct system compromise.

- **Behavioral Focus:** Modern SOC analysis relies on process and network behavior rather than static URL appearance alone.
- **Purple Team Mindset:** Understanding offensive redirection techniques directly improves the ability to configure accurate SIEM detection rules and deception traps.

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